SHA256 tag over the caveat, as shown in figure 9.6. You then throw away the old authentication tag and append the caveat and the new tag to the macaroon. Because it's infeasible to reverse HMAC to recover the old tag, nobody can remove caveats that have been added unless they have the original key.

> **WARNING** Because anybody can add a caveat to a macaroon, it is important that they are used only to restrict how a token is used. You should never trust any claims in a caveat or grant additional access based on their contents.

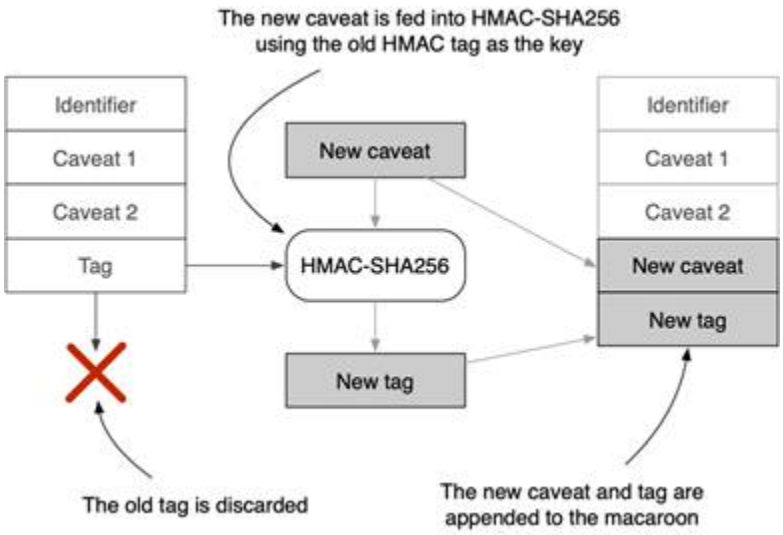


Figure 9.6 To append a new caveat to a macaroon, you use the old HMAC tag as the key to authenticate the new caveat. You then throw away the old tag and append the new caveat and tag. As nobody can reverse HMAC to calculate the old tag they cannot remove the caveat.